

Matheus Gomes Cordeiro

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SKILLS

Programming Languages: Python, R, SQL, C/C++, MATLAB/Octave, Wolfram Mathematica

Technologies & Tools: Git, GitHub, GitLab, Databricks, RabbitMQ, Elasticsearch, TensorFlow/Keras, scikit-learn, PyTorch, NLTK, spaCy, statsmodels, NeuralForecast by Nixtla, Prophet, PySpark, python-bcb, BacenAPI, sktime, NumPy, Numba, Cython, mpmath, pytest, pip, pandas, pymove, Matplotlib, Plotly, Seaborn, Voilà, Panel, Streamlit, pdoc, Transformers/Hugging Face, PostgreSQL, MongoDB, Jupyter Notebook, Google Colab, LaTeX, Markdown

PROFESSIONAL EXPERIENCE

University of Fortaleza (Unifor), Fortaleza, Brazil

Jul 2026 – Present

Assistant Professor – Part-Time

- Teach on-site undergraduate courses in Mathematics for Computing, Problem Modeling with Mathematics, and Logical and Algorithmic Reasoning.
- **Mathematics for Computing:** propositional, first-order, and predicate logic; proof methods; set theory and relations; functions, sequences, and recurrence; and counting methods, including permutations, arrangements, and combinations.
- **Problem Modeling with Mathematics:** functions and limits applied to modeling; derivatives, rates of change, Newton–Raphson, and optimization; integrals, Riemann sums, numerical integration, the Fundamental Theorem of Calculus, and geometric and computational applications.
- **Logical and Algorithmic Reasoning:** algorithm representation and development; variables, data types, operators, and input/output; conditional and repetition structures; functions and fundamental algorithms; arrays, matrices, linear and binary search, and bubble sort.
- Plan and deliver lectures, activities, and assessments, connecting mathematical and algorithmic foundations to computational problem-solving and practical programming applications.

ACT Digital

Aug 2025 – Present

Data Scientist II

- Bank Budget Forecasting:
 - Developed AI-driven forecasts of credit-loss provisions based on the monthly Expected Credit Loss percentage (ECL%) to support compliance with Central Bank of Brazil (BACEN) Resolution No. 4,966.
- Credit Model Validation Agent MVP:
 - Worked as a consultant on the development of a Minimum Viable Product (MVP) for an AI agent designed to test credit models' adherence to defined validation criteria, in collaboration with Domo, a third-party company.
- Standardized Credit Model Validation Framework:
 - Led the technical design, at Banco BMG, of a proposal for a standardized credit model validation pipeline structured as a reusable framework.
 - Designed a scoring process to quantitatively assess validation criteria and outcomes on a consistent scale, improving consistency, comparability, and traceability across evaluated models.
 - Mapped validation scores to qualitative governance outcomes — *Accepted*, *Accepted with Reservations*, or *Rejected* — using predefined score ranges to support model approval decisions.
- Tools and Methodologies:
 - Applied time-series and machine-learning models, including ARIMA/ARIMAX, SARIMA/SARIMAX, AutoARIMA, N-BEATS, N-HiTS, Autoformer, FEDformer, Prophet, KNeighborsRegressor, and XGBRegressor.
- Technologies:
 - Developed technical solutions in R (RStudio) and Python using NumPy, pandas, scikit-learn, Matplotlib, statsmodels, sktime, NeuralForecast by Nixtla, Prophet, PySpark, python-bcb, and BacenAPI.
 - Used SQL and Databricks for data processing, analysis, and management.
- Client:
 - Banco BMG.

Data Scientist II

- i) Risk Model Validation through VOYAGER IT Innovative Solutions as an outsourced consultant:
 - Validated four models covering Credit Risk, Microcredit Risk, Operational Risk, Market Risk, and Social, Environmental, and Climate Risks.
 - Ensured model compliance and governance by reviewing documentation against internal policies, external regulations, and governance frameworks.
 - Performed advanced technical analyses, including:
 - Exploratory Data Analysis (EDA) to identify relevant patterns and insights.
 - Data preprocessing to improve the quality and reliability of the information used by the models.
 - Reproduction of existing models to verify the accuracy and consistency of their results.
 - Development of challenger models to test and improve predictive performance.
 - Model performance assessment for benchmarking and continuous improvement.
- ii) Tools and Methodologies:
 - Applied supervised machine-learning algorithms, including MLP, XGBoost, LightGBM, and XGBSE, as well as unsupervised techniques such as K-means, PAM, and CLARA.
 - Used statistical modeling methods, including Value at Risk (VaR), rule-based systems, and simplified mathematical models such as RAROC.
- iii) Developed technical solutions in R (RStudio), Python using NumPy, pandas, scikit-learn, Matplotlib, and TensorFlow, and SQL.

Quantum Optics and Information Group (GIQ), Fortaleza, Brazil

Feb 2023 – Feb 2025

Quantum Computing Simulation Researcher

- Conducted research on the accurate and efficient simulation of the Quantum Harmonic Oscillator, a widely used model in photonic quantum computing.
The research resulted in the development of a Python package named **Fast Wave**. The work was co-advised by Dr. Scott Glancy from the National Institute of Standards and Technology (NIST):
Fast Wave GitHub Repository ,
Fast Wave Documentation .
- Python 3, Numba, Cython, mpmath, pytest, pdoc, pip, Google Colab, Git, and GitHub.

SeaFortress, Fortaleza, Brazil

Jan 2018 – Present

Chief Artificial Intelligence Officer – CAIO

- Co-founder and Chief Artificial Intelligence Officer of **SeaFortress**, a startup operating under a hybrid business model that combines an independent game studio with a software development company.
Led the application of Artificial Intelligence across independent game development and custom virtual experiences, focusing on accessible and cost-efficient solutions.
Contributed directly to **Sentis**, Unity's deep-learning inference library, and had a SeaFortress game showcased at **GDC 2024**:
SeaFortress Website .
- Python 3, Transformers/Hugging Face, PyTorch, C#, Sentis, and Unity.

Insight Data Science Lab, Fortaleza, Brazil

Jan 2022 – Jan 2023

Artificial Intelligence Researcher

- Conducted research combining Natural Language Processing (NLP) and vehicle trajectory analysis.
Trained BERT-based models from the Hugging Face Transformers library on vehicle trajectory datasets to assess whether the next trajectory element could be predicted similarly to next-word prediction in a text sequence.
The research resulted in the development of a dashboard application named **Teach**:
Teach GitHub Repository .
- Python 3, Transformers/Hugging Face, PyTorch, pandas, pymove, Voilà, Jupyter Notebook, and GitHub.

LEAD Dell, Fortaleza, Brazil

Jul 2021 – Jan 2022

Artificial Intelligence Developer

- Worked as part of a Research and Development team responsible for building a machine-learning application to detect anomalies in Dell systems through system logs.
Integrated the Isolation Forest model from scikit-learn into a pipeline that received logs from RabbitMQ, processed the data, and stored the results in Elasticsearch.
- Python 3, scikit-learn, RabbitMQ, and Elasticsearch.

EDUCATION

Federal University of Ceará (UFC), Fortaleza, Brazil

Feb 2023 – Feb 2025

Master of Science in Teleinformatics Engineering

Main research areas: Scientific Computing, Computational Physics, Quantum Computing Simulation, Quantum Devices and Systems, and Quantum Optics.

Federal University of Ceará (UFC), Fortaleza, Brazil

Jan 2017 – Jan 2023

Bachelor of Science in Computer Engineering

Relevant coursework: Linear Algebra; Differential, Integral, Vector, and Numerical Calculus; Ordinary Differential Equations; Probability and Statistics; Data Mining; Programming Logic; Object-Oriented Programming; Data Structures; Image Processing; Graph Algorithms; Graph Theory; and Neural Networks.

PUBLICATIONS

- [1] Cordeiro, M., Bezerra, I. P., & Vasconcelos, H. H. M. (2024). Efficient computation of the wave function $\psi_n(x)$ using Hermite coefficient matrix in Python. In *Proceedings of the 7th Workshop-School on Quantum Computing and Information (7th WECIQ)* (pp. 56–60). Rio de Janeiro: CEFET/RJ.
- [2] Cordeiro, M., Markert, C., Araújo, S. S., Campos, N. G. S., Gondim, R. S., da Silva, T. L. C., & da Rocha, A. R. (2022). Towards smart farming: Fog-enabled intelligent irrigation system using deep neural networks. *Future Generation Computer Systems*, 129, 115–124. <https://doi.org/10.1016/j.future.2021.11.013>
- [3] Cordeiro, M. G., Serafim, P. B. S., Nogueira, Y. L. B., Vidal, C. A., & Cavalcante Neto, J. B. (2019). A minimal training strategy to play Flappy Bird indefinitely with NEAT. In *2019 18th Brazilian Symposium on Computer Games and Digital Entertainment (SBGames)* (pp. 21–28). <https://doi.org/10.1109/SBGames.2019.00014>
- [4] Cruz, L. A., Coelho da Silva, T. L., Magalhães, R. P., Melo, W. C. D., Cordeiro, M., de Macedo, J. A. F., & Zeitouni, K. (2022). Modeling Trajectories Obtained from External Sensors for Location Prediction via NLP Approaches. *Sensors*, 22(19), 7475. <https://doi.org/10.3390/s22197475>
- [5] Freitas, N. A. de, da Silva, T. C., de Macêdo, J. F., Melo Junior, L., & Cordeiro, M. (2021). Using deep learning for trajectory classification. In *Proceedings of the 13th International Conference on Agents and Artificial Intelligence – Volume 2: ICAART* (pp. 664–671). SciTePress. <https://doi.org/10.5220/0010227906640671>

CERTIFICATIONS

Learning Tracks:

- Data Analysis in Databricks — DataCamp

Professional Courses:

- Data Management in Databricks — DataCamp
- Databricks Concepts — DataCamp
- Introduction to Databricks — DataCamp
- Data Visualization in Databricks — DataCamp
- Credit Risk Modeling in Python — DataCamp

Professional Accreditations:

- Academy Accreditation – Databricks Fundamentals
- NVIDIA Fundamentals of Deep Learning

Academic and Scientific Credentials:

- Certificate of Publication — Sensors
- Certificate of Publication — Future Generation Computer Systems